A Periodic Table of the Finite Simple Groups

Every finite simple group appears exactly once, in its simplest form. Columns are grouped by rank; the vertical axis is the field size q.

TYPE

🟧 Cyclic groups · 🟪 Alternating groups · 🟦 Chevalley groups · 🟩 Steinberg groups · 🟨 Suzuki and Ree groups · 🟪 Sporadic groups

HOW TO READ A CELL

$\cong A_2(2)$ — other descriptions of the same group
$A_1(7)$ — the group
$2^3\cdot3\cdot7$ — prime factorisation of its order

AN EMPTY CELL

the group is not defined there, or is not simple,
or it is isomorphic to one printed elsewhere —
every group appears only under its simplest name:

$A_2(1) \to C_3$
$A_1(2^2), A_1(5) \to A_5\ [A_4(1)]$
$A_1(3^2) \to A_6\ [A_5(1)]$
$A_2(2) \to A_1(7)$
$A_3(2) \to A_8\ [A_7(1)]$
$^2A_3(2) \to B_2(3)$
$C_n(q)$, q even $\to B_n(q)$
$B_2(2)' \to A_6\ [A_5(1)]$
$G_2(2)' \to {}^2A_2(3)$
$^2G_2(3)' \to A_1(2^3)$

FAMILIES

$A_n(q) = L_{n+1}(q)$ — Linear
$B_n(q) = O_{2n+1}(q)$ — Orthogonal (odd)
$C_n(q) = S_{2n}(q)$ — Symplectic
$D_n(q) = O^+_{2n}(q)$ — Orthogonal (plus)
$G_2(q)$ — Exceptional G_2
$F_4(q)$ — Exceptional F_4
$E_6(q)$ — Exceptional E_6
$E_7(q)$ — Exceptional E_7
$E_8(q)$ — Exceptional E_8
$^2A_n(q) = U_{n+1}(q)$ — Unitary
$^2D_n(q) = O^-_{2n}(q)$ — Orthogonal (minus)
$^3D_4(q)$ — Triality
$^2E_6(q)$ — Twisted E_6
$^2B_2(q)$ — Suzuki
$^2G_2(q)$ — Ree (small)
$^2F_4(q)$ — Ree (large)

q	Group	Order
q	C_p	p
q	$A_1(q)$	$q(q^2-1)/\gcd(2,q-1)$
q	$A_2(q)$	$q^3(q^3-1)(q^2-1)/\gcd(3,q-1)$
q	$B_2(q)$	$q^4(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$G_2(q)$	$q^6(q^6-1)(q^2-1)$
q	$^2A_2(q)$	$q^3(q^3+1)(q^2-1)/\gcd(3,q+1)$
q	$^2B_2(2^{2m+1})$	$2^{4m+2}(2^{4m+2}+1)(2^{2m+1}-1)$
q	$^2G_2(3^{2m+1})$	$3^{6m+3}(3^{6m+3}+1)(3^{2m+1}-1)$
q	$A_3(q)$	$q^6(q^4-1)(q^3-1)(q^2-1)/\gcd(4,q-1)$
q	$B_3(q)$	$q^9(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_3(q)$	$q^9(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$^2A_3(q)$	$q^6(q^4-1)(q^3+1)(q^2-1)/\gcd(4,q+1)$
q	$A_4(q)$	$q^{10}(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(5,q-1)$
q	$B_4(q)$	$q^{16}(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_4(q)$	$q^{16}(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$D_4(q)$	$q^{12}(q^6-1)(q^4-1)(q^4-1)(q^2-1)/\gcd(4,q^4-1)$
q	$F_4(q)$	$q^{24}(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)$
q	$^2A_4(q)$	$q^{10}(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(5,q+1)$
q	$^2D_4(q)$	$q^{12}(q^6-1)(q^4+1)(q^4-1)(q^2-1)/\gcd(4,q^4+1)$
q	$^3D_4(q)$	$q^{12}(q^8+q^4+1)(q^6-1)(q^2-1)$
q	$^2F_4(2^{2m+1})$	$2^{24m+12}(2^{12m+6}+1)(2^{8m+4}-1)(2^{6m+3}+1)(2^{2m+1}-1)$
q	$A_5(q)$	$q^{15}(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(6,q-1)$
q	$B_5(q)$	$q^{25}(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_5(q)$	$q^{25}(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$D_5(q)$	$q^{20}(q^8-1)(q^6-1)(q^5-1)(q^4-1)(q^2-1)/\gcd(4,q^5-1)$
q	$^2A_5(q)$	$q^{15}(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(6,q+1)$
q	$^2D_5(q)$	$q^{20}(q^8-1)(q^6-1)(q^5+1)(q^4-1)(q^2-1)/\gcd(4,q^5+1)$
q	$A_6(q)$	$q^{21}(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(7,q-1)$
q	$B_6(q)$	$q^{36}(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_6(q)$	$q^{36}(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$D_6(q)$	$q^{30}(q^{10}-1)(q^8-1)(q^6-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^6-1)$
q	$E_6(q)$	$q^{36}(q^{12}-1)(q^9-1)(q^8-1)(q^6-1)(q^5-1)(q^2-1)/\gcd(3,q-1)$
q	$^2A_6(q)$	$q^{21}(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(7,q+1)$
q	$^2D_6(q)$	$q^{30}(q^{10}-1)(q^8-1)(q^6+1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^6+1)$
q	$^2E_6(q)$	$q^{36}(q^{12}-1)(q^9+1)(q^8-1)(q^6-1)(q^5+1)(q^2-1)/\gcd(3,q+1)$
q	$A_7(q)$	$q^{28}(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(8,q-1)$
q	$B_7(q)$	$q^{49}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_7(q)$	$q^{49}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$D_7(q)$	$q^{42}(q^{12}-1)(q^{10}-1)(q^8-1)(q^7-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^7-1)$
q	$E_7(q)$	$q^{63}(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^2-1)/\gcd(2,q-1)$
q	$^2A_7(q)$	$q^{28}(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(8,q+1)$
q	$^2D_7(q)$	$q^{42}(q^{12}-1)(q^{10}-1)(q^8-1)(q^7+1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^7+1)$
q	$A_8(q)$	$q^{36}(q^9-1)(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(9,q-1)$
q	$B_8(q)$	$q^{64}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_8(q)$	$q^{64}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$D_8(q)$	$q^{56}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^8-1)$
q	$E_8(q)$	$q^{120}(q^{30}-1)(q^{24}-1)(q^{20}-1)(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^2-1)$
q	$^2A_8(q)$	$q^{36}(q^9+1)(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(9,q+1)$
q	$^2D_8(q)$	$q^{56}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8+1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^8+1)$
q	$A_9(q)$	$q^{45}(q^{10}-1)(q^9-1)(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(10,q-1)$
q	$B_9(q)$	$q^{81}(q^{18}-1)(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$C_9(q)$	$q^{81}(q^{18}-1)(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
q	$D_9(q)$	$q^{72}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^9-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^9-1)$
q	$^2A_9(q)$	$q^{45}(q^{10}-1)(q^9+1)(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(10,q+1)$
q	$^2D_9(q)$	$q^{72}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^9+1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^9+1)$
q	$A_n(q)$	$q^{n(n+1)/2}\prod_{i=2}^{n+1}(q^i-1)/\gcd(n+1,q-1)$
q	$B_n(q)$	$q^{n^2}\prod_{i=1}^{n}(q^{2i}-1)/\gcd(2,q-1)$
q	$C_n(q)$	$q^{n^2}\prod_{i=1}^{n}(q^{2i}-1)/\gcd(2,q-1)$
q	$D_n(q)$	$q^{n(n-1)}(q^n-1)\prod_{i=1}^{n-1}(q^{2i}-1)/\gcd(4,q^n-1)$
q	$^2A_n(q)$	$q^{n(n+1)/2}\prod_{i=2}^{n+1}(q^i-(-1)^i)/\gcd(n+1,q+1)$
q	$^2D_n(q)$	$q^{n(n-1)}(q^n+1)\prod_{i=1}^{n-1}(q^{2i}-1)/\gcd(4,q^n+1)$

Sporadic groups

Name	Group	Order
Mathieu	M_{11}	$2^4\cdot3^2\cdot5\cdot11$
Mathieu	M_{12}	$2^6\cdot3^3\cdot5\cdot11$
Mathieu	M_{22}	$2^7\cdot3^2\cdot5\cdot7\cdot11$
Mathieu	M_{23}	$2^7\cdot3^2\cdot5\cdot7\cdot11\cdot23$
Mathieu	M_{24}	$2^{10}\cdot3^3\cdot5\cdot7\cdot11\cdot23$
Janko	J_1	$2^3\cdot3\cdot5\cdot7\cdot11\cdot19$
Hall–Janko	J_2	$2^7\cdot3^3\cdot5^2\cdot7$
Janko	J_3	$2^7\cdot3^5\cdot5\cdot17\cdot19$
Janko	J_4	$2^{21}\cdot3^3\cdot5\cdot7\cdot11^3\cdot23\cdot29\cdot31\cdot37\cdot43$
Higman–Sims	HS	$2^9\cdot3^2\cdot5^3\cdot7\cdot11$
McLaughlin	McL	$2^7\cdot3^6\cdot5^3\cdot7\cdot11$
Held	He	$2^{10}\cdot3^3\cdot5^2\cdot7^3\cdot17$
Rudvalis	Ru	$2^{14}\cdot3^3\cdot5^3\cdot7\cdot13\cdot29$
Suzuki	Suz	$2^{13}\cdot3^7\cdot5^2\cdot7\cdot11\cdot13$
O'Nan	O′N	$2^9\cdot3^4\cdot5\cdot7^3\cdot11\cdot19\cdot31$
Conway	Co_1	$2^{21}\cdot3^9\cdot5^4\cdot7^2\cdot11\cdot13\cdot23$
Conway	Co_2	$2^{18}\cdot3^6\cdot5^3\cdot7\cdot11\cdot23$
Conway	Co_3	$2^{10}\cdot3^7\cdot5^3\cdot7\cdot11\cdot23$
Fischer	Fi_{22}	$2^{17}\cdot3^9\cdot5^2\cdot7\cdot11\cdot13$
Fischer	Fi_{23}	$2^{18}\cdot3^{13}\cdot5^2\cdot7\cdot11\cdot13\cdot17\cdot23$
Fischer	Fi_{24}'	$2^{21}\cdot3^{16}\cdot5^2\cdot7^3\cdot11\cdot13\cdot17\cdot23\cdot29$
Harada–Norton	HN	$2^{14}\cdot3^6\cdot5^6\cdot7\cdot11\cdot19$
Lyons	Ly	$2^8\cdot3^7\cdot5^6\cdot7\cdot11\cdot31\cdot37\cdot67$
Thompson	Th	$2^{15}\cdot3^{10}\cdot5^3\cdot7^2\cdot13\cdot19\cdot31$
Baby Monster	B	$2^{41}\cdot3^{13}\cdot5^6\cdot7^2\cdot11\cdot13\cdot17\cdot19\cdot23\cdot31\cdot47$
Monster	M	$2^{46}\cdot3^{20}\cdot5^9\cdot7^6\cdot11^2\cdot13^3\cdot17\cdot19\cdot23\cdot29\cdot31\cdot41\cdot47\cdot59\cdot71$

CLASSIFICATION THEOREM

Every finite simple group is cyclic of prime order, alternating of degree n ≥ 5, a group of Lie type, or one of the 26 sporadic groups. This poster shows every one of them, each exactly once.

NOTES

- After the cyclic column the columns are grouped by rank: every family with a simple group of rank 1, then of rank 2, and so on.
- E_8 is the last family to appear, so from rank 9 onwards every block is the same six classical families — hence the ellipsis.
- Row q = 1: over “the field with one element” a group of Lie type degenerates to its Weyl group — for type A_n the symmetric group S_{n+1} — whose simple part is A_{n+1}, just as the simple part of $GL_{n+1}(q)$ is $A_n(q) = L_{n+1}(q)$. Only type A is simple there, so that row is exactly the alternating groups.
- “$A_n(1)$” is a position on this table, not a received notation: there is no field with one element, and the q → 1 limit is a heuristic of Tits’. Each group in that row is printed under its own name, A_{n+1}.
- A cell carrying a parameter shows the generic order formula in place of a prime factorisation, and so does a cell whose factorisation is too long to print.
- $C_n(q) \cong B_n(q)$ for q even, so only the odd-q cells of the C columns are printed.
- $^2B_2(q)$ and $^2F_4(q)$ exist for $q = 2^{2m+1}$ and $^2G_2(q)$ for $q = 3^{2m+1}$ (m ≥ 1); their other rows are empty.
- $^2F_4(2)$ is not simple, but its derived subgroup — the Tits group — is, and is printed in that cell.
- The 26 sporadic groups belong to no family of Lie type; they are shown in their own strip.
- Above each column is the Dynkin diagram of its family. On the Steinberg, Suzuki and Ree columns the red arrows are the automorphism of the diagram that the Frobenius twist uses — for $^3D_4$ a turn of order three about the branch node.
- A symbol with a subscript and no field is a group over “F_1”: C_p is cyclic of order p and A_m is alternating of degree m. With a field, $A_n(q)$ is linear and $C_n(q)$ symplectic — the argument tells them apart.